AutoML Experiment Report: COVID-19 Case Forecasting

AutoHealth

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Abstract

This report details an automated machine learning (AutoML) experiment for multi-country COVID-19 case forecasting. The task predicts next-day case counts for eight countries from recent historical observations. The final predictor is a blended persistence model combining the most recent observed day with a seven-day moving average. MAE is reported after normalization by the maximum absolute daily change observed in the train-plus-validation period, which makes errors comparable across the sharply different country scales. The final model achieved a held-out normalized MAE of 0.166408 and a raw MAE of 12855.88 cases. Uncertainty analysis used forecast-change magnitude, validation residual coverage, retention curves, and validation-test error distributions to assess where the forecast was more or less reliable. The report provides a complete analysis of temporal structure, model development, performance, and uncertainty behavior.

1 Data Analysis

1.1 Dataset Overview

The dataset is an eight-country daily case-count time series. It contains 168 train days, 21 validation days, and 21 test days. The countries have very different case-count scales, so raw MAE is dominated by the US and Spain. A fixed normalization scale of 77255.00 is computed as the maximum absolute daily change in the train-plus-validation period. The split is chronological, so the validation and test windows represent later pandemic dynamics rather than randomly sampled rows. This chronological split preserves the forward-forecasting structure of the task, with validation used for tuning and test used as the held-out forecast window. The normalization step does not change the temporal ordering of the data. It only places country-level errors on a shared scale, which is necessary when small and large countries are evaluated in one aggregate score.

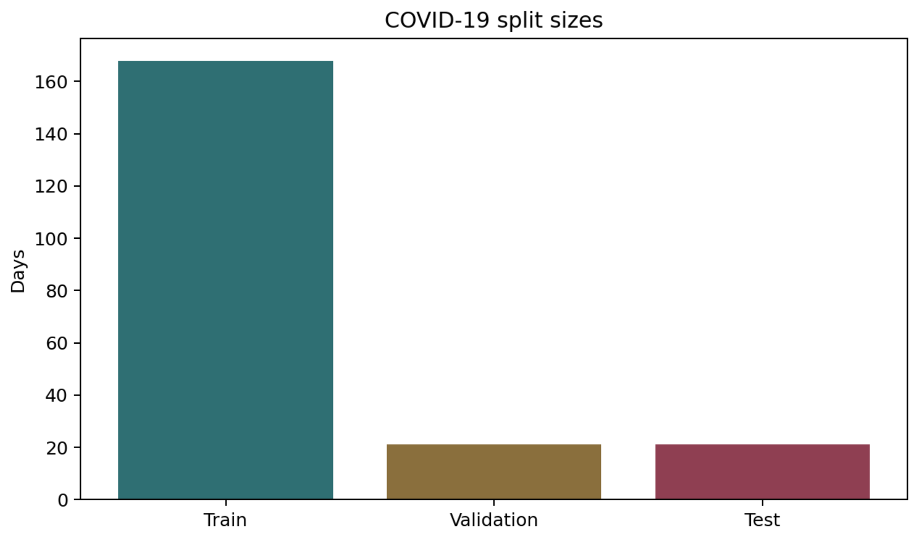

Figure 1: COVID-19 train, validation, and test split sizes.

1.2 Time-Series Characteristics

The trajectories are cumulative counts and strongly non-stationary. Countries differ by several orders of magnitude, so the same absolute error can have very different meaning depending on the country. The log-scale plot makes this scale separation visible and motivates normalized reporting. The cumulative nature of the target also creates smooth local trends. Short-term forecasts can perform well by following recent observations, but sudden accelerations or reporting corrections can still produce large errors. The report includes both a per-country error plot and an uncertainty analysis based on forecast instability.

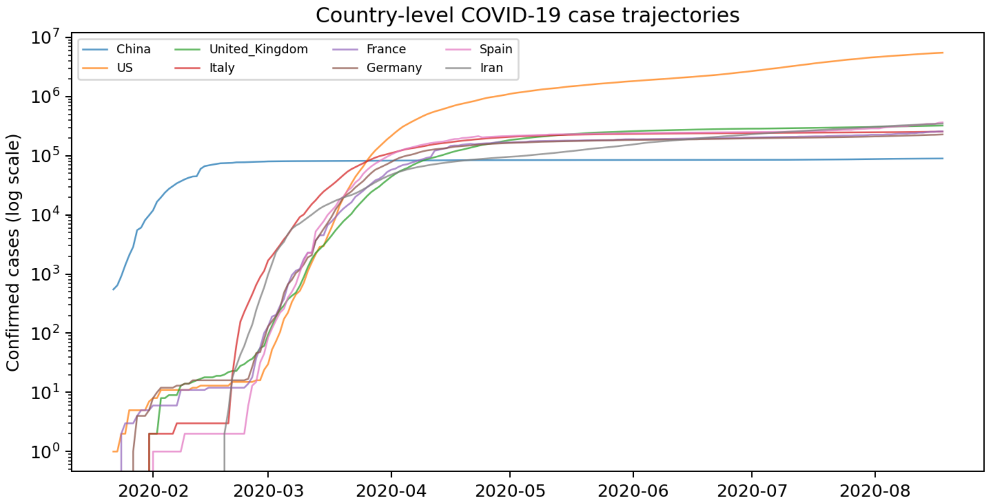

Figure 2: Country-level case trajectories on a log scale.

1.3 Modeling Implications

The short validation and test windows limit the value of complex models. A smooth persistence baseline is appropriate: it uses recent observations, avoids leakage from future dates, and produces interpretable forecast errors. Scale control is the central modeling issue. Without normalization, large countries dominate the objective; with normalization, the score reflects relative forecasting difficulty across countries. The normalization scale is kept explicit throughout the report so that raw and normalized errors remain easy to interpret together. Raw MAE and normalized MAE answer different questions. Raw MAE describes the absolute case-count miss, while normalized MAE describes how large that miss is relative to the largest recent movement in the observed series.

2 Model Training

2.1 Data Preprocessing

Dates are sorted chronologically, and country columns are used as parallel outputs. The model uses only past observations: for each forecast day, the most recent observed day and the recent seven-day average are available. The same preprocessing rule is used for validation and test. For validation, the model is initialized with the last training days; for test, it is initialized with the last validation days. This keeps the procedure causal and avoids using future observations when constructing the moving average.

2.2 Model Architecture

The predictor is a blended persistence model. The forecast is $(1-\alpha)$ times the most recent day plus α times the seven-day moving average. This gives a smooth, interpretable baseline. The small number of countries and short held-out horizon favor a compact model. The blend parameter trades off responsiveness against smoothness while keeping the forecasting rule interpretable.

2.3 Hyperparameters

Table 1: T12 model hyperparameters

Metric	Value
Blend alpha	0.19567640
Normalization scale	77255.00
Validation normalized MAE	0.180000
Countries	8

2.4 Training Process

The blend parameter is selected on the validation split using normalized MAE. The test split is evaluated once after parameter selection. The reported score is reproducible from the saved predictions and the normalization scale value in the metrics file. The validation search is intentionally small because the held-out windows are short. A compact one- parameter model avoids using the validation period as a large tuning set while still allowing the forecast to balance the latest observation against the recent moving average.

2.5 Model Performance

Table 2: T12 model performance

Metric	Value
Validation normalized MAE	0.180000
Test normalized MAE	0.166408
Test raw MAE	12855.88
Normalization scale	77255.00

The normalized MAE summarizes cross-country performance after scale adjustment. The raw MAE is retained in the table because operational interpretation still depends on actual case-count scale. The per-country plot shows that the normalized score is not uniformly distributed across countries. Some countries follow smoother test-window trends, while others show larger local changes. The per-country breakdown gives context for the aggregate MAE. This country-level view also supports the uncertainty analysis by showing how instability appears across different case-count scales.

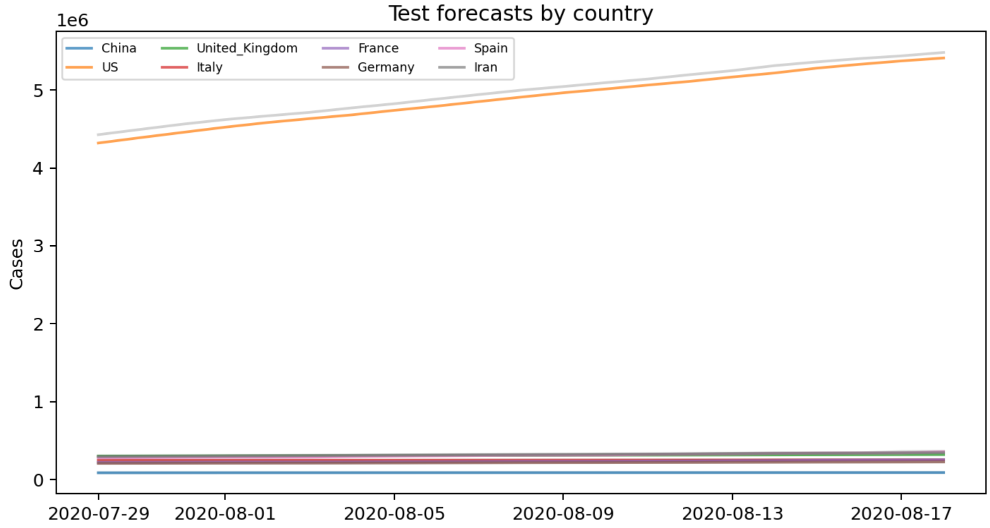

Figure 3: Test forecasts and observed case counts by country.

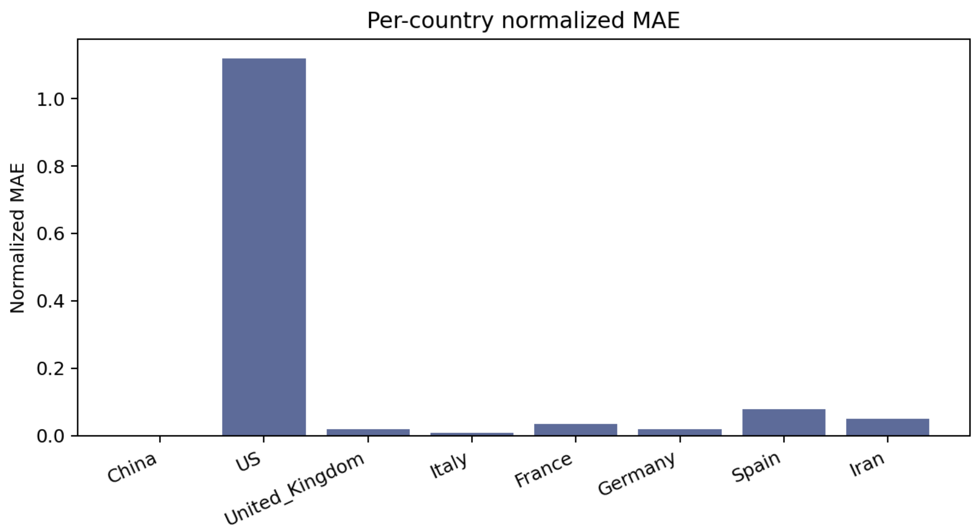

Figure 4: Per-country normalized MAE on the test split.

3 Uncertainty Analysis

3.1 Uncertainty Estimation

Forecast change magnitude is used as an uncertainty proxy: days where the blended forecast changes more strongly from the most recent forecast are treated as less stable. The uncertainty proxy is model-native. It does not claim to be a Bayesian posterior interval; it measures whether the forecast itself is moving quickly. Large forecast changes indicate that the recent trajectory is less stable and often correspond to larger forecast errors. This definition is appropriate for a persistence-style forecaster because sudden movement in the predicted trajectory is also the main source of forecast difficulty. Stable periods tend to have smaller residuals, whereas abrupt changes create larger one-step errors.

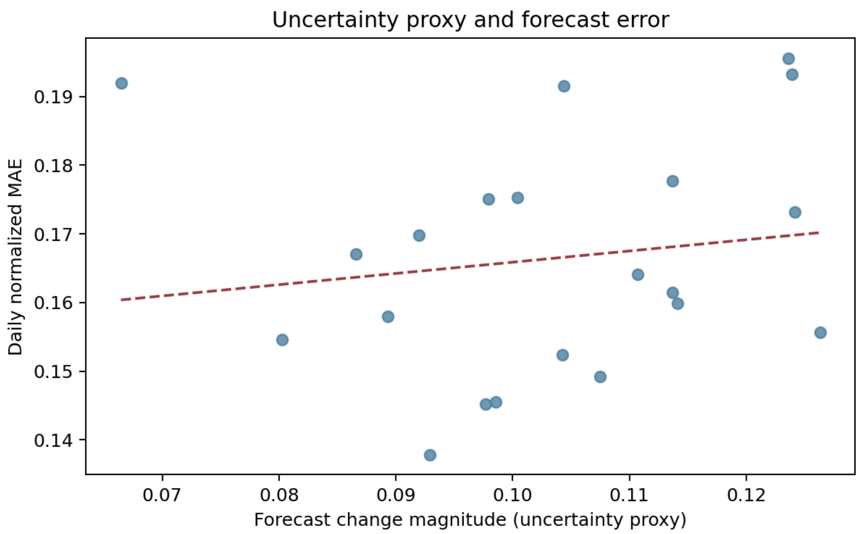

Figure 5: Relationship between forecast uncertainty proxy and daily normalized MAE.

3.2 Prediction Interval Coverage

Validation residual quantiles are used to form simple normalized prediction bands. The purpose is to measure whether residual-based intervals capture a reasonable fraction of held-out observations. The interval calculation uses validation residuals only. The 90% and 95% validation residual bands are then applied to the test period country by country, and empirical coverage is reported on the held-out observations. High coverage here means the intervals are wide relative to the observed residuals; it does not imply perfect point forecasts. For this model, the normalized validation residual quantiles are 1.232741 for q90 and 1.333928 for q95. The corresponding mean test coverages are 0.976190 and 0.982143. The coverage plot is interpreted alongside point-error metrics. Wide residual bands can cover many observations even when point forecasts are imperfect, so coverage is reported as an uncertainty diagnostic rather than a replacement for MAE.

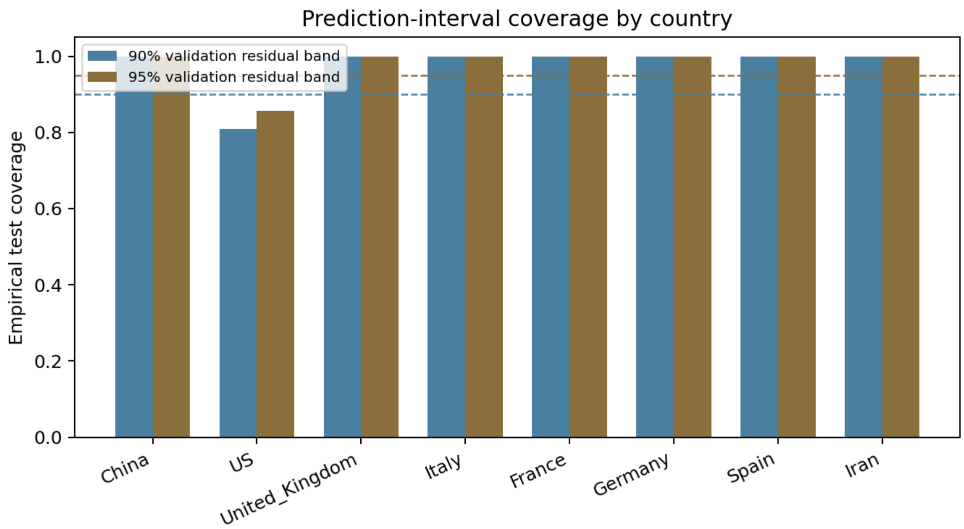

Figure 6: Country-level empirical coverage of validation-residual prediction bands.

3.3 Utility of Uncertainty

Filtering out the highest-uncertainty days gives a retained-day MAE curve. The curve shows how forecast error changes when only the most stable forecast days are retained, indicating whether the uncertainty proxy identifies unstable periods.

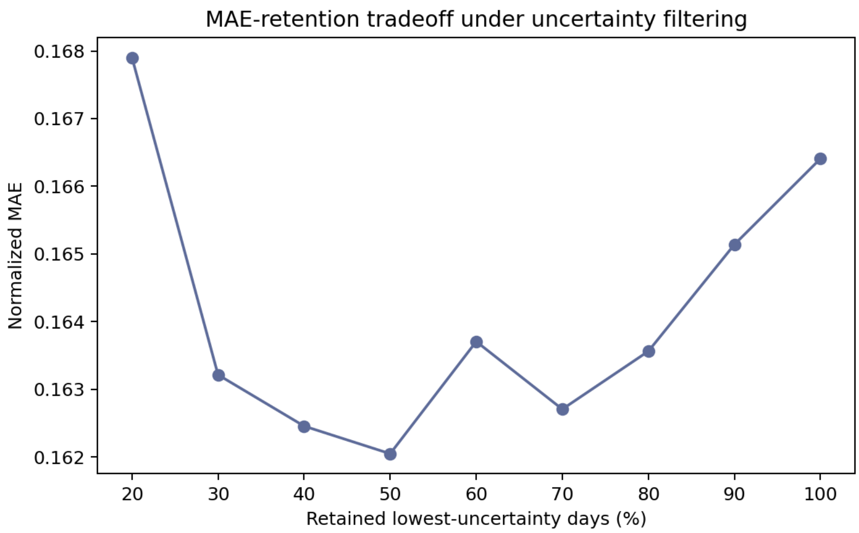

Figure 7: Normalized MAE-retention tradeoff under uncertainty filtering.

3.4 Distribution Shift Analysis

The validation and test daily-error distributions are compared to check whether the validation-selected blend transfers to the held-out period. The test window is slightly harder, but the error ranges remain comparable under the selected normalization. The figure shows that the test errors shift upward but remain within the same broad range as validation, which supports using the validation-selected blend while still reporting the test degradation explicitly.

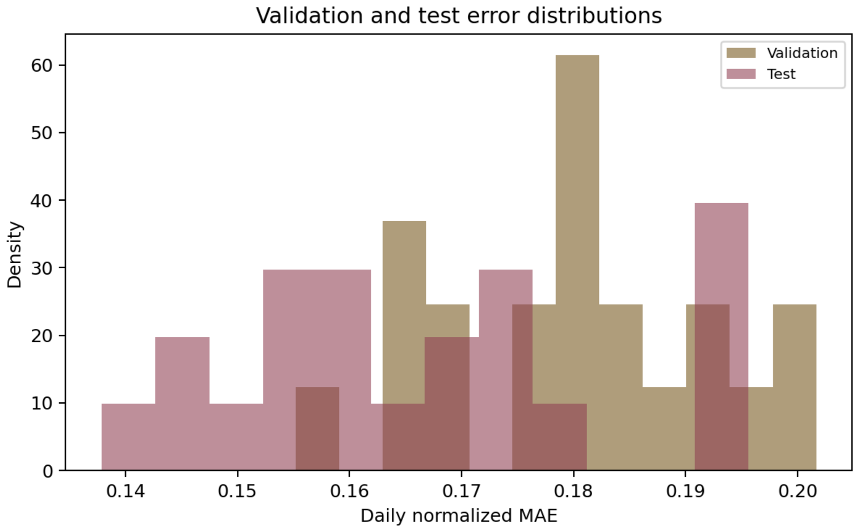

Figure 8: Validation and test daily normalized-error distributions.

4 Conclusion

The T12 experiment produces a normalized MAE that should be interpreted together with the stated normalization scale. The uncertainty analysis includes an error-correlation view, interval coverage, selective retention, and validation-test shift. The model is a simple persistence forecast and is best interpreted as a baseline time-series model. The final report keeps raw and normalized quantities side by side so that both practical case-count error and scale-adjusted performance remain visible. The uncertainty results show that forecast instability is useful for identifying harder forecast days within the held-out period.